MOTOROLA MICROSYSTEMS

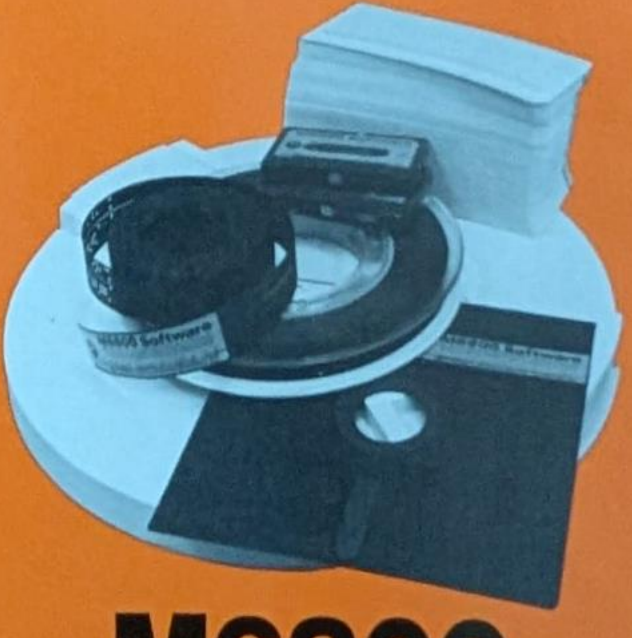
The use of microprocessors can greatly simplify many hardware designs. However, a microprocessor-based design includes software, an element not found in conventional hard-wired designs. The proper tools are essential to the design of a microprocessor system.

Although microprocessor software can be written directly in machine language, this is too tedious, time-consuming, error-prone, and expensive for all but the smallest system.

Most microprocessor software is presently written in assembly language. At this level, the programmer has complete and explicit control of the microprocessor, while the assembler relieves him of many burdensome details. High-level languages (compilers) are becoming increasingly popular for programming microprocessors. Although such programs are typically somewhat larger and slower than those written in assembly language, they can be developed more rapidly and maintained more easily.

Motorola provides software development tools for three environments:

1. **Resident Software** – Use of the EXORciser for software development permits dual use of this system development equipment. This low-cost alternative for software development employs the same prototyping hardware used for program test and hardware/software integration. Software development tools also are available for Motorola's Evaluation Modules.
2. **In-House Computer Systems** – Cross-computer software provides low-cost program development and simulation facilities for in-house computer systems ranging from minicomputers to large mainframe machines. Capabilities of an in-house system may include powerful system software, high-speed peripherals, and multiple-user access.
3. **Time-Sharing** – The user has immediate worldwide access to M6800 software development packages. Time-sharing provides access to a powerful user-oriented system at a minimal initial cost. It may also be used in meeting workload peaks and critical schedules.



M6800 Support Software